

Mouloud Mammeri University of Tizi-Ouzou

Department of Computer Science

Module: English for Computer Science

Session 4: The Internet and Networking

Session Objective

The aim of this session is to introduce students to basic computer and Internet vocabulary, understand the difference between **data** and **information**, recognize key computer acronyms, and develop awareness of cybersecurity and online privacy.

Data vs Information

Data is raw fact, symbol, or measurement that have not yet been processed or given meaning. They represent the input to a system.

Information is the result of processing, organizing, or interpreting data in a way that makes it meaningful and useful for decision-making or communication.

Term	Definition	Example
Data	Raw, unorganized facts (numbers, symbols, text).	25°C, 10.5 MB, “Nassima”, 01/11/2025.
Information	Processed data that conveys meaning or context.	“The temperature today is 25°C, warmer than yesterday.”

Note: *Data* refers to input facts, while *Information* represents useful output derived from those facts.

Computer Acronyms

Before we start, remember that an **acronym** is a short form made from the initial letters of a group of words (for example, “CPU” for “Central Processing Unit”).

Acronym	Meaning	Example
CPU	Central Processing Unit	The CPU executes the computer's instructions.
RAM	Random Access Memory	Adding more RAM improves performance.
ROM	Read Only Memory	ROM stores permanent system data.
GPU	Graphics Processing Unit	A powerful GPU is needed for gaming and AI tasks.
SSD	Solid State Drive	SSDs are faster than traditional hard drives.
HDD	Hard Disk Drive	HDDs are used for large data storage.
BIOS	Basic Input/Output System	The BIOS runs when you start your computer.
OS	Operating System	Linux is an open-source OS.
GUI	Graphical User Interface	Windows has a user-friendly GUI.
CLI	Command Line Interface	Linux administrators often use the CLI.
PDF	Portable Document Format	Please send the report as a PDF file.
API	Application Programming Interface	The app connects to the server using an API.
DBMS	Database Management System	MySQL is a popular DBMS.
AI	Artificial Intelligence	AI allows machines to learn and make decisions.
ML	Machine Learning	ML helps systems improve automatically from experience.

Internet Vocabulary

Let's learn some common words related to the Internet and networking. Each term is followed by its English definition and its French equivalent (in parentheses).

- **Internet:** A global network connecting millions of computers around the world. (*Internet*)
- **Website:** A collection of related web pages stored under one domain name. (*Site web*)

- **Web Address :** The unique address of a website on the Internet.
- **Browser:** A program used to access and view websites, such as Chrome or Firefox.
(*Navigateur*)
- **Link / Hyperlink:** A clickable text or image that takes you to another webpage.
(*Lien / Hyperlien*)
- **Email:** A method of exchanging digital messages over the Internet. (*Courriel / E-mail*)
- **Search Engine:** A website that helps users find information online, such as Google.
(*Moteur de recherche*)
- **Download:** To copy data from the Internet to your device. (*Télécharger*)
- **Upload:** To send data from your device to the Internet. (*Téléverser*)
- **Cloud:** Online storage space where you can keep your files and access them anywhere. (*Nuage / Cloud*)
- **Social Media:** Websites or apps that allow users to share content and connect with others. (*Réseaux sociaux*)

Internet and Network Acronyms

Acronym	Meaning	Example
IP	Internet Protocol	Every device on the Internet has a unique IP address.
IPv4 / IPv6	Internet Protocol version 4 / 6	IPv6 provides more IP addresses than IPv4.
MAC	Media Access Control	Each device has a unique MAC address.
LAN	Local Area Network	All computers in the office are connected to a LAN.
WAN	Wide Area Network	The Internet is the largest WAN in the world.
ISP	Internet Service Provider	My ISP offers high-speed fiber Internet.
DNS	Domain Name System	DNS translates domain names into IP addresses.
HTTP / HTTPS	Hypertext Transfer Protocol / Secure	Secure websites begin with HTTPS.
FTP	File Transfer Protocol	FTP is used to upload files to a web server.
SMTP	Simple Mail Transfer Protocol	SMTP is used to send emails.
URL	Uniform Resource Locator	The URL of Wikipedia is www.wikipedia.org .
WWW	World Wide Web	The WWW connects billions of websites.
VPN	Virtual Private Network	A VPN protects your privacy on public Wi-Fi.
Wi-Fi	Wireless Fidelity	I connected my laptop to the Wi-Fi network.
IoT	Internet of Things	The IoT connects smart devices like sensors and cameras.
IPsec	Internet Protocol Security	IPsec secures communication over the Internet.

Cybersecurity and Online Privacy

In today's digital world, protecting our personal information and online activities is very important. Cybersecurity and privacy help us stay safe while using the Internet and prevent hackers or malicious software from accessing our data.

Term	Definition	Example
Cybersecurity	Protection of systems from attacks or unauthorized access.	Cybersecurity prevents hackers from stealing data.
Privacy	Keeping personal information safe online.	Online privacy is essential for everyone.
Password	A secret word to access an account.	Use a strong password with numbers and symbols.
Virus	Malicious software that harms your system.	A virus can destroy important files.
VPN	A secure network that encrypts your data.	I use a VPN when connected to public Wi-Fi.
Online Safety	Safe behavior on the Internet.	Online safety means being careful on social media.

Most Common Cyber Threats

1. Malware

Malware (short for *malicious software*) is any **program designed to harm your computer or steal your data**. It can delete files, slow down your system, or spy on your activity. **Almost every cyberattack involves some type of malware.**

2. Ransomware

A **type of malware** that **locks your data or device** and **demands money (a ransom)** to unlock it. Hackers usually ask for payment in cryptocurrency. **Example: Your files become inaccessible until you pay.**

3. Trojan Horse

A program that **looks safe or useful**, but actually **contains harmful code inside**. It tricks users into installing it and then gives hackers access to the system. **Example: A fake game or security app hiding a virus.**

4. Spyware

Spyware is software that secretly collects information about you — like your passwords or bank details — and sends it to hackers.

It runs quietly in the background without your knowledge.

5. Phishing

A scam using fake emails, websites, or messages to trick people into revealing personal information or passwords.

Example: An email pretending to be from your bank asking you to “confirm your account”.

Note:

Cyberattacks can come from cybercriminals, hackers, or even insider threats. Awareness and good security practices help reduce risks.

Best Practices for Online Security

To stay safe online, follow these basic cybersecurity practices:

- Use complex passwords and change them regularly.
- Enable two-factor authentication (2FA) for extra protection.
- Avoid public Wi-Fi for sensitive activities.
- Don't share personal details on social networks.
- Regularly update your software and antivirus.